

Overall design

The project implements MVC design, separates servlet, model, repository, and service.

Uses Hibernate for database interactions.

JSP components

```
<c:forEach var="conversation" items="${allCon}">
    <div class="query">
        <p>${conversation.userQuery}</p>
    </div>

    <div class="response">
        <p>${conversation.chatbotResponse}</p>
    </div>
</c:forEach>
```

The JSP page uses a forEach to generate all conversations from a list. Then display variables in each object in the list.

Chatbot logic

The chatbot uses a predefined HashMap to generate response. When user's input includes a keyword that matches the keys in the map, it will return the corresponding response.

The method is in the service.

Database schema

Only one table is used.

Conversations	
PK	<u>id</u>
	user_query
	chatbot_response
	timestamp

Functionality

After user input a question, the app will match it with the predefined map. Then it will store the question and answer in database, and extract all conversations in the database and use EL to display in the screen.

Screenshot

